A light ray AO from the point object is incident on the mirror and it is reflected along OB. The normal is ON.

The angle of incidence $\angle AON$ = angle of reflection $\angle BON$

Another ray AD incident normally on the mirror at D is reflected back along DA. When BO and AD are extended backwards, they meet at a point A'. Thus, the rays appear to come from a point A' which is behind the plane mirror. The object and its image are at equal perpendicular distances from the plane mirror which can be shown by the following explanation.

Angle $\angle AON$ = angle $\angle DAO$ [Since they are alternate angles]

Angle $\angle BON$ = angle $\angle OA'D$ [Since they are corresponding angles]

Hence, it follows that angle, $\angle DAO = \angle OA'D$

The triangles ΔODA and $\Delta ODA'$ are congruent

$$\therefore AD = A'D$$

This shows that the image distance d_i inside the plane mirror is equal to the object distance d_o in front of the plane mirror.

The image formed by the plane mirror for extended object is shown in Figure 6.4(b).

6.1.5 Characteristics of the image formed by plane mirror

(i) The image formed by a plane mirror is virtual, erect and laterally inverted sidewise (left / right).

(ii) The size of the image is equal to the size of the object.

(iii) The image distance behind the mirror is equal to the object distance in front of the mirror.

(iv) If an object is placed between two plane mirrors inclined at an angle θ, then the number of images n formed is given in Table 6.1 the images formed are shown in Figure 6.5.

Table 6.1 Image by inclined mirrors

$\left(\frac{360}{\theta}\right)$	The position of object placed	Number of images n
Even	Symmetrical	$n = \left(\frac{360}{\theta} - 1\right)$
	Unsymmetrical	$n = \left(\frac{360}{\theta} - 1\right)$
Odd	Symmetrical	$n = \left(\frac{360}{\theta} - 1\right)$
	Unsymmetrical	$n = \left(\frac{360}{\theta}\right)$



Figure 6.5 Images formed by inclined mirrors

EXAMPLE 6.2

What is the height of the mirror needed for a person to see his/her image fully on the mirror?

Solution

Let us assume a person of height h is standing in front of a vertical plane mirror. The person could see his/her head when light from the head falls on the mirror and gets reflected to the eyes. Same way, light from the feet falls on the mirror and gets reflected to the eyes.

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